

Appendix B to the Georgia Growth Model Report Application of the SIMEX Method for Measurement Error Correction to the Georgia Growth Model

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Abstract

This Appendix to Georgia Student Growth Model Technical Reports provides methodological details about the SIMEX method for correcting for measurement error in assessment data. Measurement error can lead to bias in statistical models and student growth estimates.

1 The Measurement Error Problem and Corrections

Measurement error (ME) is an inherent component of all standardized tests, and the impact that ME can have when test score data are used to compute student growth and teacher/school evaluation measures has been the focus of a growing body of academic research. Specifically in the area of Student Growth Percentile (SGP) measures of student progress, ME has been found to create bias that can disadvantage students with lower prior achievement and vice versa. This bias is transferred to aggregate measures of educator effectiveness when a disproportionate number of students with relatively low/high prior achievement are concentrated in a classroom or school (Lockwood & Castellano, 2015; Shang, VanIwaarden, & Betebenner, 2015).

Researchers have proposed several useful methods for correcting the effects of ME. The use of Simulation/Extrapolation (SIMEX) techniques has been found to effectively eliminate the ME induced bias related to prior achievement in SGPs (Shang et al., 2015), and this method is currently available for SGP calculation in the [SGP package](#) (Betebenner, VanIwaarden, Domingue, & Shang, 2017) for the [R statistical program](#). Currently several states utilize the SIMEX corrected measures in their student growth modeling and evaluation policies.

2 The SIMEX Method

The SIMEX method was proposed by Cook and Stefanski (1994) as a measurement error (ME) correction technique when the standard error of measurement (SEM) is known or can be reasonably well estimated. The SIMEX method is a functional approach that does not make strong assumptions about variable distributions (Battauz, Bellio, & Gori, 2011). Compared with other methods, SIMEX is much easier to implement for measurement error models that are less understood, such as that involving nonparametric quantile regression (QR). It relies on repeated random sampling to solve the problem, similar to bootstrap or jackknife, hence its simplicity and generality (L. Stefanski & Cook, 1995). For a detailed description and discussion of SIMEX see Carroll, Ruppert, Stefanski, & Crainiceanu (2006).

The basic idea of the method is to gauge the dependence of the ME effect on SEM through a series of experiments. Increasing amounts of simulated ME are added to observed values, and results from these experiments are then used to extrapolate the relationship of interest to the point where SEM is equal to zero. To explain further, let σ_{ui}^2 stand for the variance of the ME term, u_i . In the simulation phase, additional ME with known variance is generated and added to the observed test scores, w_i , to create increasingly error-prone “pseudo” data sets and then “pseudo” parameter estimates in the following steps. First, choose a set of monotonically increasing small numbers, denoted as λ . For example, let $\lambda = 0.5, 1, 1.5, 2$. Then, for each value of λ , produce an artificial error $\sqrt{\lambda}v_i$, where v_i is randomly generated from the distribution of u_i . The inflated ME, $u_i + \sqrt{\lambda}v_i$, would have a variance equal to $(1 + \lambda)\sigma_{ui}^2$. Next, the “pseudo” data sets which are contaminated with the inflated ME are used to produce the “pseudo” parameter estimates with the chosen statistical model. In order to reduce sampling noise, the simulation and “pseudo” estimation are repeated for B times, and the sample mean of the B “pseudo” parameter estimates is calculated at each given λ . In the extrapolation stage, the averaged “pseudo” parameter estimates and the “naive” estimates (the original estimates obtained from the unperturbed data) are regressed on λ . Finally, when λ is set to be equal

to -1, the predicted value of the extrapolant function would be the SIMEX estimate of the error-free parameter.

In the SGP model, the interest lies in estimating $\widehat{SGP}_X$, and these quantities are derived from the fitted values of the model, not its regression coefficients. Following the example of Carroll et. al. (1999), the SIMEX process described above is carried out on the fitted values: “pseudo” fitted value estimates, $\hat{Q}_W^{(\tau)}(\lambda, b)$, for each of the $\tau = 1, 2, \dots, 99$ percentiles are obtained with the repeatedly perturbed “pseudo” data sets. These values are averaged over B at each λ , regressed on λ , and finally extrapolated to $\lambda = -1$ to produce the SIMEX estimate $\hat{Q}_{(X, SIMEX)}^{(\tau)}$. In the case of quantile crossing, $\hat{Q}_{(X, SIMEX)}^{(\tau)}$ is sorted at the specific x_i , as recommended in Dette and Volgushev (2008) and Chernozhukov, Fernandez-Val and Glichon (2010).

The choices of λ , B , and extrapolation function demand explanations. Various authors provided rules-of-thumb (Carroll et al., 2006, etc.; see, for example, L. Stefanski & Cook, 1995). The commonly adopted values for λ are a few equally spaced numbers between 0 and 2; B is usually fixed at 100; and the extrapolant function is often specified to be linear, quadratic, or non-linear regressions. We conducted Monte Carlo experiments to compare linear with quadratic extrapolants under various λ specifications. Our results show that the linear extrapolation is generally a better choice than the quadratic. With very fine λ grid, such as $\lambda = 0, 2/25, 4/25, \dots, 50/25$, the quadratic SIMEX estimator of SGP is slightly less biased than the linear one, but, with a much larger variability, its MSE is still considerably higher than that of the linear estimator. As for the choice of λ , a finer grid significantly improves the quadratic estimator but makes little difference for the linear one. The MSE of the SIMEX estimator decreases monotonically as B increases, but the return diminishes for $B > 30$. The detailed results are omitted. For additional information on the use of the SIMEX method to correct for covariate measurement error, see Shang, VanIwaarden and Betebenner (2015).

3 The Effect of the SIMEX Method on Individual SGPs

The following sections examine the effect of the SIMEX method on individual SGP estimates and aggregations of them at the class/school level respectively. Results from several simulation studies are presented, followed by a discussion section. Findings from the application of the SIMEX method to Georgia 2012 assessment data and practical issues to consider based for production level application of SIMEX are also provided.

The performance of the SIMEX method with the SGP model is tested in the following steps. Scale scores in `sgpData` of the `SGP` package (a toy dataset extracted from longitudinal assessment data) were treated as the “true scores” for a two cohorts of students. The first cohort consists of 6,977 4th grade students with a single 3rd grade prior scale score available. The other consists of 6,468 5th grade students with 4th and 3rd grade priors scores available. Missing values were excluded. Normally distributed measurement errors were then generated using the conditional standard errors of measurement of the various grades provided in `SGPstateData` in the same package. The “observed” scores are the sum of the “true” scores and the generated measurement errors. SGP analysis on the “true” scores and the SGP with SIMEX analysis on the “observed” scores are then respectively to produce “true” SGPs, “observed” (or “naive”) SGPs, and SIMEX corrected SGPs. To minimize sampling errors, 100 “observed” data sets

were generated using the above method. Each student ultimately has a true SGP, 100 observed SGPs from the 100 generated data sets, and 100 SIMEX SGPs from the same data sets.

3.1 Quantifying the Performance of the SIMEX Method

Bias and Mean Squared Error (MSE) are the usual ways of quantifying the performance of statistical methods. In a simulation study where the major outcome is the estimation of a few model parameters, both are usually straight forward to calculate—bias is the difference between the estimated and the true parameter averaged over simulation replications, and MSE is the squared difference between the estimated and the true parameter averaged over simulation replications.

In the simulation study of the SGP analysis, however, each replication of the analysis produces results for thousands of students. To evaluate the estimators, a way to summarize across both students and replications is needed. The following statistics to compare SGPs estimated with and without SIMEX are derived for this purpose:

$$Total\ MSE = \frac{1}{n} \sum_{i=1}^n \frac{1}{R} \sum_{r=1}^R (\widehat{SGP}_{i,r} - SGP_i)^2 \quad (1)$$

Where $i = 1, 2, \dots, n$ indicates students, $r = 1, 2, \dots, R$ indicates simulation replications, $\widehat{SGP}_{i,r}$ denotes the estimated SGP (either with or without SIMEX) for student i in replication r , and SGP_i denotes the “true” SGP for student i .

$$Mean\ Bias = \frac{1}{n} \sum_{i=1}^n \frac{1}{R} \sum_{r=1}^R (\widehat{SGP}_{i,r} - SGP_i) \quad (2)$$

$$SD\ Bias = Standard\ Deviation_{across\ i} \left(\frac{1}{R} \sum_{r=1}^R \widehat{SGP}_{i,r} - SGP_i \right) \quad (3)$$

Total MSE is a comprehensive standard which accounts for both bias and variability. *Mean Bias* is the bias averaged over students and replications, but since *Mean Bias* of either estimator is likely to be very close to 0, we also calculated *SD Bias*, which is derived by taking the difference between the estimated and the true SGPs averaged across replications, and then calculate the standard deviation of these differences across students. *SD Bias* may be a better indicator of the magnitude of bias than *Mean Bias*, just like Standard Error of Measurement is a better indicator of the magnitude of measurement errors than the mean, which is usually 0.

3.2 Results

The following table presents the results of the simulation study.

\caption*{\textbf{Table B.1**}}

SGP Without SIMEX			SGP With SIMEX		
<i>Total MSE</i>	<i>Mean Bias</i>	<i>SD Bias</i>	<i>Total MSE</i>	<i>Mean Bias</i>	<i>SD Bias</i>

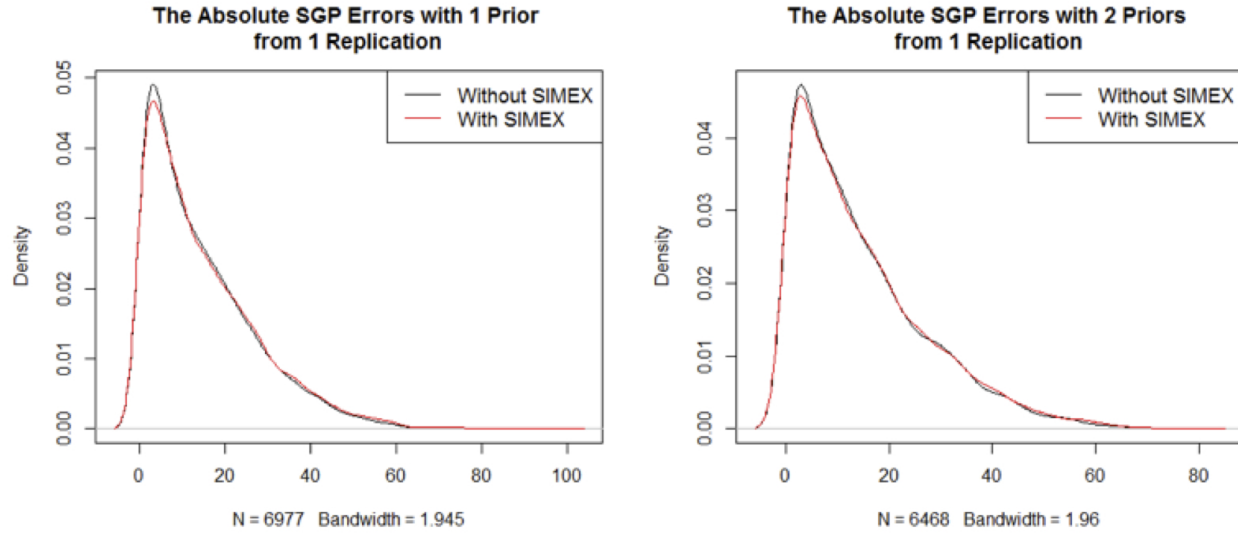


Figure 1: **Fig. B.1** *Absolute values of SGP Errors with 1 and 2 Priors Used*

	SGP	Without	SIMEX		SGP With	SIMEX
Gr 4 ~ Gr 3	362.980	0.019	8.004	393.244	-0.107	6.612
Gr 5 ~ Gr 4, 3	360.795	-0.001	7.512	379.277	0.088	6.602

3.3 Why Are the Total MSEs So Large?

The total MSEs are all very large. The square roots of the MSEs, generally considered a measure of the average magnitude of errors, are around 19, far greater than the *Mean Bias* or the *SD Bias*.

Figure B.1 shows the density of the absolute values of the SGP (with and without SIMEX) errors with 1 and 2 prior year scores respectively from a single replication of the simulation study. Plots of other replications are highly similar. Figure B.1 shows that, in any given replication of the simulation study, the absolute differences between estimated and true SGPs are highly skewed with a thin but long tail to the right. A small number of the errors in SGPs estimated with 1 prior are close to 100. Among SGPs estimated with 2 priors, it seems that there are less extreme outliers. SGPs estimated with SIMEX seem to mirror those without SIMEX closely, although the former tend to have more very large errors. The relatively small number of outliers can have dramatic influence on the MSE because, when squared, they become *much* larger and the skewness of the distributions of the squared errors will dramatically increase compared with those in Figure B.1.

3.4 Comparing the SIMEX Results with the Non-SIMEX Results

Due to the skewness discussed in the section above, the root mean square error is not a good measure of the magnitude of errors in this context. The results of the SGP analysis with

and without SIMEX are therefore based on two criteria—*SD Bias* and *MSE*. The comparison is straightforward: the SIMEX results have smaller biases but larger MSEs which indicates larger variability across replications. Furthermore it seems that in both scenarios (i.e. with 1 and 2 priors) the increase of variability outweighs the reduction of bias.

There is, however, something strange here. In earlier studies of the SIMEX method for correcting biased model coefficients, we noticed much bigger bias reductions than those presented in Table B.1. The SIMEX estimators do generally have larger variability, but usually this drawback is outweighed by the prominent gains in bias reduction. Why is the SIMEX method not doing well this time?

3.5 Why Doesn't the SIMEX Method Perform Well in Correcting Individual SGP?

The question about the validity of using the SIMEX method to correct individual level SGPs can be raised with the following rationale. Suppose that the model $\hat{y}_{GOOD} = a + bx_{OBSERVED}$ is estimated when $x_{OBSERVED}$ is measured with errors, but $\hat{y}_{BETTER} = \alpha + \beta x_{TRUE}$ is the actual model desired. Now the SIMEX method, or any other measurement error correction, moves us from a and b closer to α and β , and often times the goal is achieved at this stage. But in the SGP analysis, simply obtaining α and β is not enough. To get more accurate SGPs, we need to obtain $\hat{y}_{BETTER}$, which is not obtainable unless we have x_{TRUE} .

Therefore, when estimating individual SGPs with SIMEX, what is roughly estimated is $\alpha + \beta x_{TRUE}$, which is neither $\hat{y}_{GOOD}$ nor $\hat{y}_{BETTER}$. Thus some bias reduction is achieved because the observed scores are usually not far from the true scores, but the effect is much discounted.

To test this theory, the “true” scores were plugged into the SIMEX process. Table B.2 presents the outcomes of SGP without SIMEX and with SIMEX combined with “true” scores from 100 simulation replications.

\caption*{\textbf{Table B.2**}}

	SGP	Without	SIMEX		SGP With SIMEX	and “True” S
	<i>Total MSE</i>	<i>Mean Bias</i>	<i>SD Bias</i>	<i>Total MSE</i>	<i>Mean Bias</i>	<i>SD</i>
Gr 4 ~ Gr 3	362.980	0.019	8.004	248.836	-0.428	
Gr 5 ~ Gr 4, 3	360.795	-0.001	7.512	287.078	-0.057	

In Table B.2, the SIMEX method greatly reduces both bias and MSE when used in combination with true scores. This shows that the fundamental reason for increased MSE in Table B.1 is the use of SIMEX in combination with observed scores.

3.6 Can We Use Estimated True Scores?

Results in Table B.2 are obviously not obtainable in reality. As a possible alternative, the estimated true scores were substituted:

$$X_{Est.True} = \text{Reliability} \times (X_{OBSERVED} - \bar{X}_{OBSERVED}) + \bar{X}_{OBSERVED} \quad (4)$$

The results obtained are almost exactly the same as those in Table B.1, which suggests that the inaccessibility of the true scores is a problem that is insurmountable without measurement replications. The transformation of the observed scores to estimated true scores does not change the rank order of the observations, and so their use in the SIMEX model yields very similar results to the use of observed scores in that model.

3.7 Visual representations of the SIMEX method corrections

Visualizations that highlight the impact of implementing the SIMEX method may provide a more intuitive understanding of the SIMEX SGP corrections and how they are obtained. Figure B.2 shows true and observed score distributions with quantile regression models fit to those two distributions. The simulated data points are bivariate normally distributed with homoscedastic, classical error structure in the prior year data. As expected, the slopes of quantile regression lines fit to the observed data are attenuated. Another notable feature is that the separation between the True 10th and 90th percentile lines is narrower than that between the corresponding lines fit to the observed data.

The SIMEX method attempts to use quantile regression to produce a “map” of the observed score distribution that more closely resembles the map of the unobserved, True score map. If the observed data (grey circles in Figure B.2) were fit with the True quantile regression models (green lines), a disproportionately large amount of high SGPs would be expected at the lower tail of the prior score distribution and, conversely, more low SGPs at the upper end of the prior score distribution. Additionally, the compressed quantile lines in the SIMEX model of the distribution would likely create fewer than expected SGPs in the middle percentile ranges. This would result in clustering of SGPs where the model cannot differentiate between similar (but not identical) score histories.

These two expected features are present in Figure B.3, which is a goodness of fit plot produced from a 4th grade Math SIMEX simulation. These plots obtained from the simulations conducted for this report suggest that the SIMEX method has produced a “better” map of what the True conditional distribution *may* look like.

3.8 Baseline Referenced SGPs

In addition to the issues discussed above for individual cohort SGPs, there are some theoretical issues that should be taken into account for the decision to use SIMEX corrections for baseline referenced SGPs. An ideal baseline model would show perfect fit (10 percent of kids in each cell of the left hand table in Figure 3). From this baseline model we hope to find evidence of system wide improvement when future scores are fit to it. That is, the baseline-referenced model can be seen as a predictive model in which we use past *observed* score distributions to predict the distribution of current *observed* scores. Unlike predictive models that attempt to predict future observations with the greatest accuracy possible, we hope that fitting the current data with this model will show misfit due to systemic changes in the student population. In particular, we hope to see higher growth in the present than what was expected (typical) in the past. This would be show up as red cells on the right side of the fit table in Figure B.3. Starting from a baseline that is already shifted due to measurement error “correction” can muddle this picture.

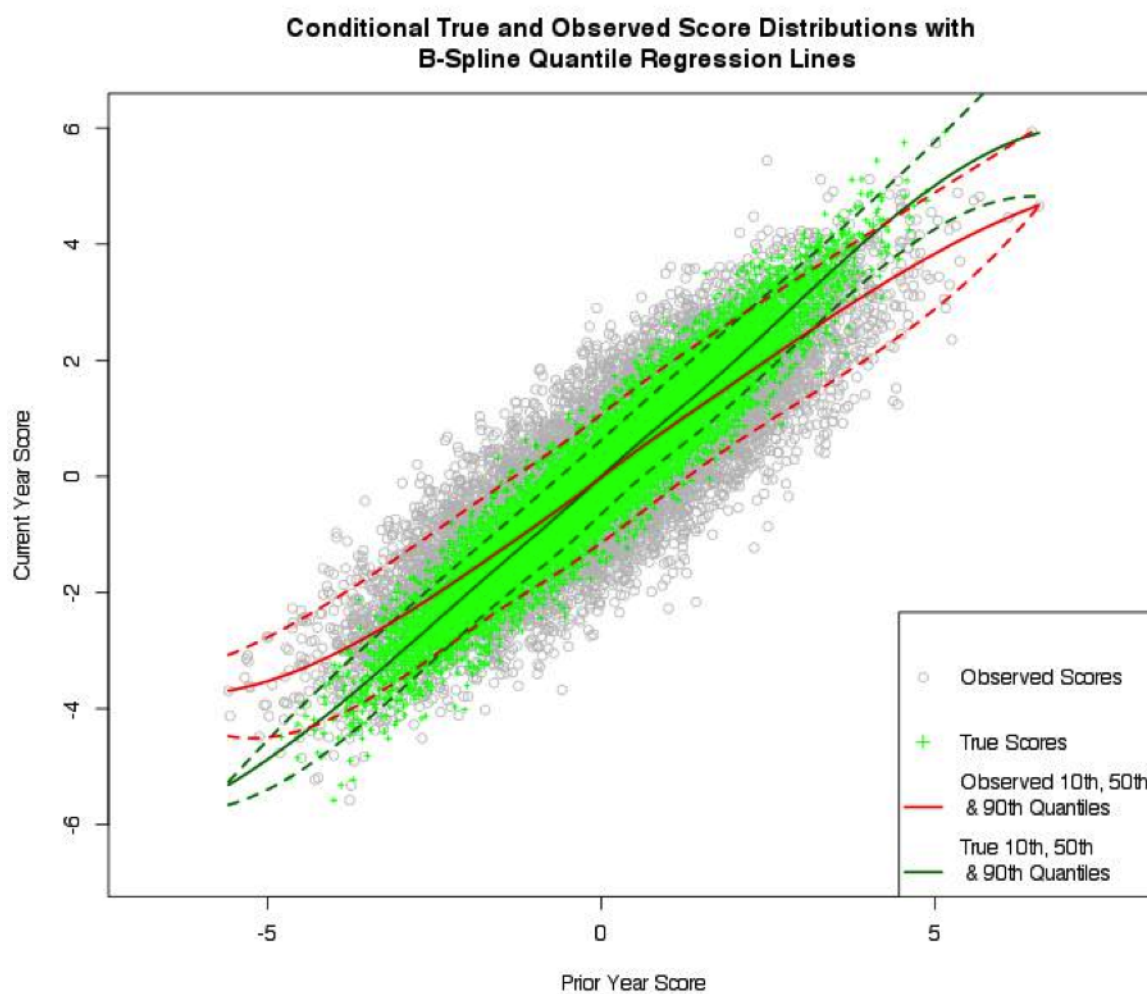


Figure 2: Fig. B.2

Student Growth Percentile Goodness-of-Fit Descriptives

2012–2013 Mathematics SGP Simex, Grade 4 (N = 4,245)

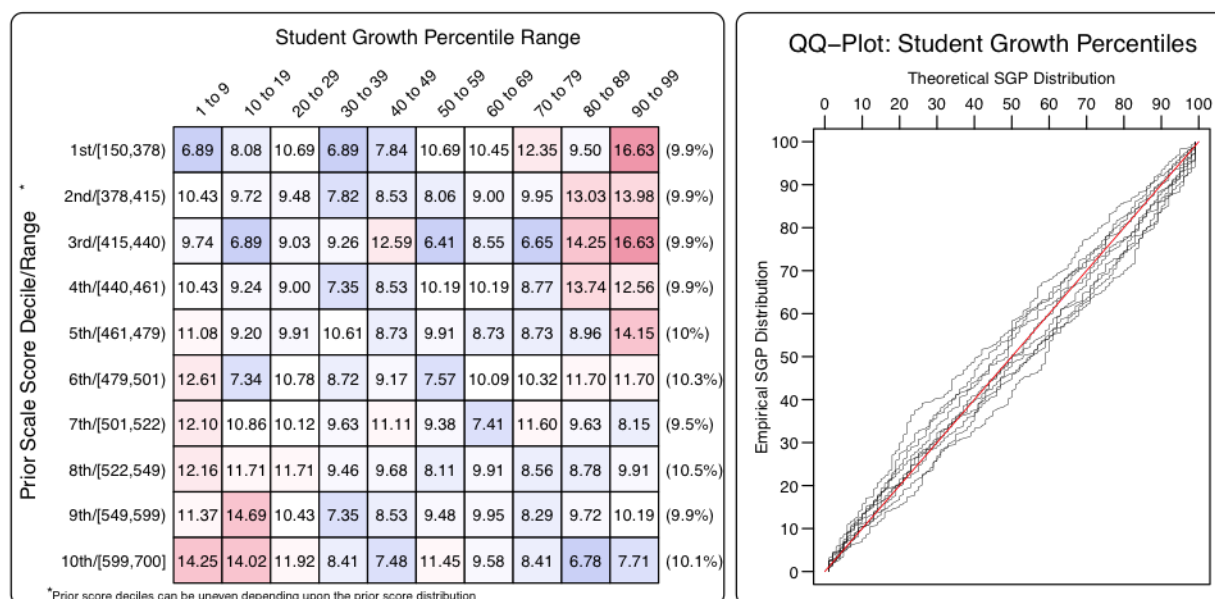


Figure 3: **Figure B.3** *Goodness of Fit plot for SIMEX SGPs obtained from one 4th grade simulation*

The measurement error literature suggests that predictive models should not use correction methods for measurement error when error prone data will be used to produce predicted values. Carroll et al. (2006, p. 38) provide this explanation,

Generally, there is no need for the modeling of measurement error to play a role in the prediction problem. If a predictor, X , is measured with error and one wants to predict a response based on the error-prone version W of X , then ... it rarely makes any sense to worry about measurement error. The reason for this is quite simple: W is an error-free as a measurement of itself! If one has an original set of data $[Y, W]$, one can fit a convenient model to Y as a function of $[W]$. Predicting Y from $[W]$ is merely a matter of using this model for prediction, that is substituting known values of W into the regression model for Y on $[W]$; the prediction errors from this model will minimize the expected squared prediction errors in the class of all linear unbiased predictors. Predictions with W naively substituted for X in the regression of Y on $[X]$ will be biased and can have large prediction errors.

In the SIMEX SGP analyses, using observed scores in place of true scores in a SIMEX corrected baseline model is equivalent to making “[p]redictions with W naively substituted for X in the regression of Y on $[X]$ ”. The large prediction errors would be evidenced in the initial non-uniform distribution of SGPs expected in the SIMEX model as shown in figure 3. These “prediction errors” would confound any information about system wide improvement (or decline) that we hope to see through model misfit as detailed above.

On the other hand, Carroll et al. (2006) also specify an exception: “The one situation requiring that we correctly model the measurement error occurs when we develop a prediction model using data from one population but we wish to predict in another population. A naive prediction model that ignores measurement error may not be transportable” (p. 39). We do not presently believe that this situation applies to the baseline-referenced analysis. We begin by assuming that current and future annual cohorts come from the baseline cohort population. If the model shows that this assumption is not reasonable, then we potentially have evidence that the system may be changing. Further research and simulations may provide some empirical evidence to clarify the issue.

4 The Effect of the SIMEX Method on Aggregated SGPs at the Class or School Level

Our conclusion that the SIMEX method should not be used to estimate individual SGPs does not automatically lead to the conclusion that the SIMEX method is not useful for correcting SGP aggregations (e.g. median growth percentiles, or MGPs), either. On the one hand, the aggregated mean/median prior scores of a class/school should be much closer to the true aggregated scores compared with individual scores, which means that the problem mentioned in preceding sections is partially solved and the SIMEX correction could be beneficial. On the other hand, the aggregated mean/median SGPs of a class/school should also be close to the true aggregated SGPs, which raises the question whether the SIMEX correction is necessary.

To investigate the utility of the SIMEX method at the school level, we examined three scenarios—randomly assigned schools, perfectly sorted schools, and actual schools which are

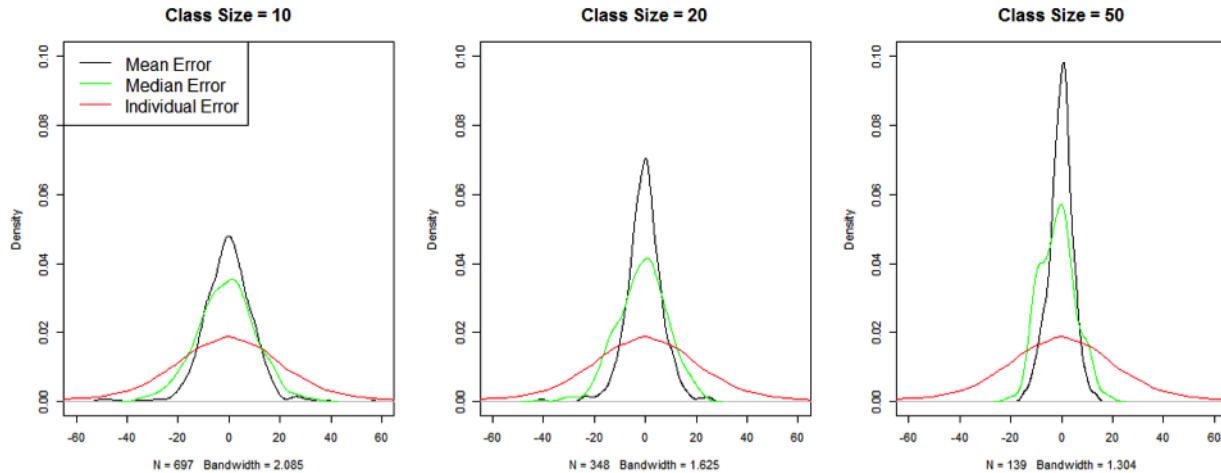


Figure 4: **Figure B.4** *Errors of Aggregated and Individual Scale Scores of Grade 3*

usually somewhere in the middle. In each scenario, we looked at the bias of aggregated scale scores and SGPs, the MSE of MGPs estimated with and without SIMEX, and the correlation between MGP (with and without SIMEX) and prior aggregated scores. From now on, we use the word “school” to refer to a collective student body, which may be a class or a school. Since we simulate schools of small and large sizes (as small as 10 students, and as large as 100), the investigation may generalize to both classes and schools.

4.1 Randomly Assigned Schools

Because measurement errors are random and independent of each other *aggregated* scale scores or SGPs are much more accurate than *individual* scale scores or SGPs. We test this claim empirically in the simulation described in the [first section of this appendix](#). Students are randomly sampled without replacement into schools of various sizes. We calculated the difference between mean observed and true scores, the difference between median observed and true scores, and the difference between individual observed and true scores. Figure B.4 plots the errors of Grade 3 scores with school sizes of 10, 20, and 50. Figure B.5 plots the errors of aggregated and individual SGPs of Grade 4. The data are drawn from a single replication of the simulation study. Plots of other grades and other replications are similar. The same plots drawn in the other scenarios, i.e. perfectly sorted schools and actual schools, are also highly similar and are not presented.

Figure B.4 and B.5 show that, when error-prone scores and SGPs are aggregated, even in schools of just 10 students, the magnitudes of errors are greatly reduced. The error reduction improves considerably as school sizes go up.

The next question is, can the SIMEX method improve the accuracy and precision of aggregated SGPs when students are randomly assigned to schools? Table B.3 presents MSE of mean and median SGPs of schools of various sizes.

\caption*{**Table B.3:** Mean Squared Errors of Aggregated SGPs of Randomized Schools of Various Sizes

Mean School SGP

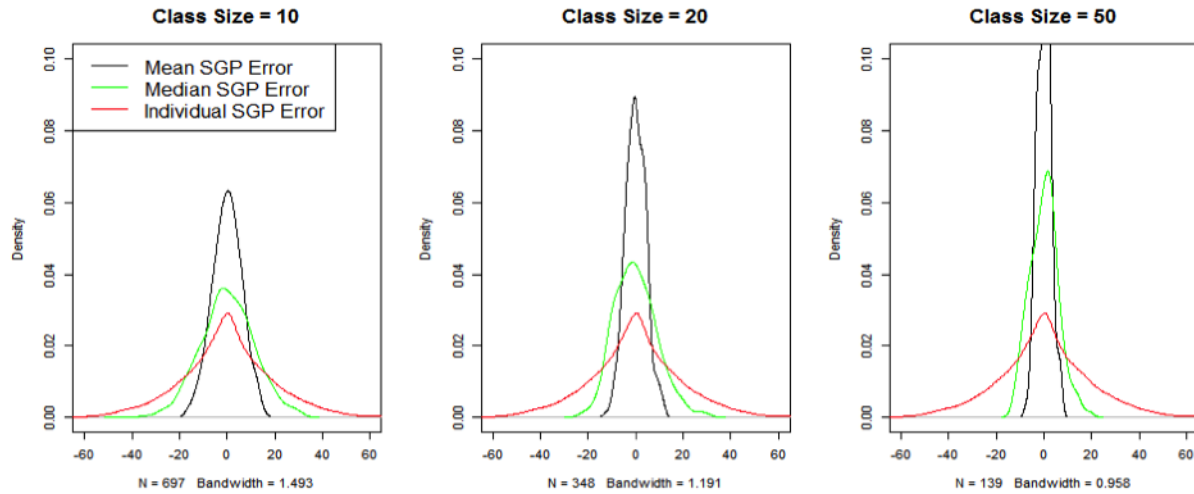


Figure 5: **Figure B.5** *Errors of Aggregated and Individual SGP's of Grade 4*

Median School SGP

MSE 10*

MSE 20

MSE 50

MSE 100

MSE 10

MSE 20

MSE 50

MSE 100

Grade 4 ~ 3

No SIMEX

36.39

17.82

7.15

3.57

131.21

83.29

39.07

18.49

SIMEX

39.29

19.45

7.79

3.96

142.2
89.92
42.82
20.48
Grade 5 ~ 4 and 3
No SIMEX
35.95
17.94
7.10
3.50

129.90
83.99
35.50
18.89
SIMEX
37.78
18.85
7.49
3.69

137.00
88.66
38.32
20.11

* *MSE 10 Means “Mean Squared Error for schools of 10 Students”*

Table B.3 shows that MSE of aggregated SGP_SIMEX are very close to and always slightly larger than MSE of non-corrected SGPs. It also shows that mean SGP seems to have much smaller MSE than median SGPs.

Lastly, we look at the correlation between aggregated SGP and prior scale scores. Table B.4 presents the correlations between aggregated observed SGP (No SIMEX) and prior scores, the correlations between SGP_SIMEX and prior scores, and the correlations between error-free SGP (true) and prior scores for schools of various sizes. Table B.4 shows that aggregated SGPs hardly correlate with aggregated prior scores in randomly assigned schools, and that the SIMEX correction tends to make the correlation estimations more erroneous rather than accurate, if only at a small scale.

\caption*{\b{Table B.4}: Correlations of Aggregated SGPs and prior scores of Randomized Schools of Various Sizes}

Correlation of *Mean* SGP and prior score

Correlation of *Median* SGP and prior score

Size 10*
Size 20
Size 50

Size 100

Size 10

Size 20

Size 50

Size 100

Grade 4 ~ 3

No SIMEX

0.061

0.018

0.014

-0.004

0.045

0.028

-0.004

0.036

SIMEX

0.040

-0.084

-0.083

-0.117

-0.029

-0.040

-0.067

-0.033

True

0.071

0.025

0.016

-0.026

0.057

0.027

0.107

-0.062

Grade 5 ~ 4 and 3

No SIMEX

0.003

0.062

0.122

0.040

-0.016

0.013	
0.104	
0.051	
SIMEX	
-0.058	
0.002	
0.066	
-0.019	
-0.062	
-0.029	
0.062	
0.020	
True	
0.003	
0.078	
0.167	
0.031	
-0.039	
0.000	
0.130	
0.059	
* Size 10 Means schools of 10 Students	

4.2 Perfectly Sorted Schools

For the second scenario, we assume that the assignment of students to different schools is completely based on the ranking of their prior true scores. Table B.5 and B.6 present the MSE and correlations calculated in this scenario similar to that of Tables B.3 and B.4. These tables show that MSE of aggregated SGPs increases dramatically when students are sorted compared with random assignment into schools. The large MSEs are somewhat mitigated when SGPs are conditioned on more than one prior scores.

Correlations between aggregated SGPs and prior scores are also greatly inflated, suggesting underestimation of MGP for under-achieving schools and overestimation of MGP at the other end. The good news is that the SIMEX method seems to be effective to a certain degree in reducing MSE, especially for larger schools. The SIMEX method also performs well in reducing the correlation between MGP and prior scores. In fact, it almost completely removes the inflation of the correlations. This confirms our earlier expectation that the SIMEX method might not be able to eliminate errors, but it can, under the right circumstances, quite effectively turn systematic bias into random errors.

\caption*{**Table B.5:** Mean Squared Errors of Aggregated SGPs of Sorted Schools of Various Sizes
Mean School SGP

Median School SGP

MSE 10*

MSE 20

MSE 50

MSE 100

MSE 10

MSE 20

MSE 50

MSE 100

Grade 4 ~ 3

No SIMEX

161.88

95.51

47.97

32.29

376.52

238.27

126.00

84.25

SIMEX

159.43

85.32

40.57

24.12

388.88

231.30

114.85

69.78

Grade 5 ~ 4 and 3

No SIMEX

161.93

81.13

36.93

25.73

350.78

206.40

97.86

67.87

SIMEX

161.87

78.91
32.44
19.32

360.49
209.31
90.24
57.81

* *MSE 10 Means "Mean Squared Error for schools of 10 Students"*

\caption*{\b{Table B.6: Correlations of Aggregated SGPs and prior scores of Sorted Schools
of Various Sizes

Correlation of *Mean* SGP and prior score

Correlation of *Median* SGP and prior score

Size 10*
Size 20
Size 50
Size 100

Size 10
Size 20
Size 50
Size 100
Grade 4 ~ 3
No SIMEX

0.305
0.420
0.577
0.645

0.263
0.364
0.515
0.602
SIMEX
0.018
0.057
0.113
0.133

0.011
0.047
0.100
0.120

True

0.005

-0.058

0.077

0.107

0.007

-0.039

0.006

0.029

Grade 5 ~ 4 and 3

No SIMEX

0.203

0.298

0.415

0.496

0.173

0.252

0.356

0.443

SIMEX

0.021

0.047

0.070

0.084

0.013

0.036

0.054

0.076

True

0.005

0.009

-0.011

-0.220

-0.015

0.005

-0.111

-0.192

* Size 10 Means schools of 10 Students

4.3 Actual Classes/Schools

Neither the randomized or perfectly sorted schools scenarios are very realistic. To get a sense of what may happen in reality, we turn to the `sgpData.LONG` dataset contained in the `SGP` R package, which is a toy dataset based on real schools in a different state. We consider scale scores in this dataset as “true” scores, and perturb them in the same way as in the previous simulations. For each perturbed data, individual and aggregated SGP and `SGP_SIMEX` are estimated. We calculated the total MSE, the Mean Bias, and the SD Bias, defined in equations (1), (2), and (3). Results from the Reading test in Grades 4 and 5 conditioning on 1 and 2 priors respectively are presented in Table B.7. Correlations of aggregated SGP and prior scores are presented in Table B.8.

\caption*{**Table B.7:** Mean Squared Errors of Aggregated SGPs of Actual Schools of Various Sizes

Mean School SGP

Median School SGP

Total MSE

Mean Bias

SD Bias

Total MSE

Mean Bias

SD Bias

Grade 4 ~ 3

Without SIMEX

5.574

-0.196

2.360

23.366

-0.104

5.137

With SIMEX

4.741

-0.075

2.177

28.364

0.034

5.329

Grade 5 ~ 4 and 3

Without SIMEX

5.948

-0.263

2.433

27.101
-0.266
5.199
With SIMEX
4.519
0.094
2.125

25.679
0.084
5.060

Table B.7 shows that MSEs of mean SGP from actual schools are quite small to begin with, and the SIMEX method is able to further reduce it to a limited extent. Bias and MSEs of median SGPs are much larger, and the effect of the SIMEX method on median SGPs seems to be mixed. Table B.8 shows that there are considerable correlations between observed aggregated SGPs and prior scores, and that even the error-free SGPs correlate significantly with prior scores. This indicates that measurement error is not the only source of the correlation. When conditioned on more than one prior score, the correlations are much reduced. When the SIMEX method is applied, the correlations are further reduced to a level that is quite close to the correlation between error-free SGPs and prior scores.

\caption*{\b{Table B.8: Correlations of Aggregated SGPs and prior scores of Actual Schools of Various Sizes

Correlation of *Mean* SGP and prior score

Correlation of *Median* SGP and prior score

Mathematics
Reading

Mathematics
Reading
Grade 4 ~ 3
No SIMEX
0.384
0.403

0.372
0.404
SIMEX
0.294
0.291

0.281
0.299

True
0.292
0.301

0.284
0.330
Grade 5 ~ 4 and 3
No SIMEX
0.220
0.256

0.172
0.252
SIMEX
0.152
0.138

0.105
0.140
True
0.159
0.121

0.093
0.119

5 Summary of Preliminary Results of Georgia SIMEX Analyses (2012)

The following sections provide summaries of the preliminary results of SIMEX analyses run using 2012 data.

5.1 *How much does the SIMEX-corrected SGP differ from non-corrected SGP?*

Answer: Not much. And when more priors are included, SGP and SGP_SIMEX appear to converge to each other, i.e. their differences have smaller means and standard deviations. The pattern is clear and consistent in Tables B.9 to B.13 which present the median, mean, and standard deviation of the SIMEX corrected SGP minus the uncorrected SGP for all students.

\caption{**Table B.9:** Reading*}

2012	Median	Mean	Std. Dev.
.....			

<i>2012</i>	Median	Mean	Std. Dev.
Gr 4, 1 prior	0	1.41	5.44
Gr 5, 1 prior	0	1.43	5.55
Gr 5, 2 priors	0	0.4	5.62
Gr 6, 1 prior	0	1.21	5.9
Gr 6, 2 priors	0	0.34	4.9
Gr 6, 3 priors	0	0.14	4.41
Gr 7, 1 prior	0	1.4	6.15
Gr 7, 2 priors	0	0.39	5.43
Gr 7, 3 priors	0	0.2	4.72
Gr 7, 4 priors	0	0.15	4.35
Gr 8, 1 prior	0	1.49	5.77
Gr 8, 2 priors	0	0.45	4.91
Gr 8, 5 priors	0	0.16	3.7

\caption{**Table B.10:** ELA*}

<i>2012</i>	Median	Mean	Std. Dev.
Gr 4, 1 prior	0	1.16	5.43
Gr 5, 2 priors	0	0.25	4.67
Gr 6, 2 priors	0	0.27	4.69
Gr 7, 2 priors	0	0.41	4.89
Gr 8, 2 priors	0	0.44	5.33

\caption{**Table B.11:** Mathematics*}

<i>2012</i>	Median	Mean	Std. Dev.
Gr 4, 1 prior	0	0.62	6.66
Gr 5, 1 prior	0	0.82	4.75
Gr 5, 2 priors	0	0.29	3.58
Gr 6, 1 prior	0	0.71	5.47
Gr 6, 2 priors	0	0.27	4.63
Gr 7, 1 prior	0	0.65	5.35

<i>2012</i>	Median	Mean	Std. Dev.
Gr 7, 2 priors	0	0.25	4.6
Gr 8, 1 prior	0	0.75	4.91
Gr 8, 2 priors	0	0.25	3.99
.....			

\caption{**Table B.12:** Science*}

<i>2012</i>	Median	Mean	Std. Dev.
.....			
Gr 4, 1 prior	0	0.48	5.13
Gr 5, 2 priors	0	0.15	4.37
Gr 6, 2 priors	0	0.1	4.17
Gr 7, 2 priors	0	0.18	4.39
Gr 8, 2 priors	0	0.15	3.87
.....			

\caption{**Table B.13:** Social Studies*}

<i>2012</i>	Median	Mean	Std. Dev.
.....			
Gr 4, 1 prior	0	0.47	4.72
Gr 5, 2 priors	0	0.11	4.31
Gr 6, 2 priors	0	0.21	3.71
Gr 7, 2 priors	0	0.15	4.2
Gr 8, 2 priors	0	0.11	3.5
.....			

5.2 *Does the SIMEX correction work differently for different schools?*

Answer: Yes. For schools with lower mean scores, the SIMEX correction tends to raise their corrected mean and median SGP, whereas for schools with higher mean scores, the SIMEX correction tends to lower their corrected mean and median SGP. This shows that SIMEX is correcting the attenuation created by measurement errors.

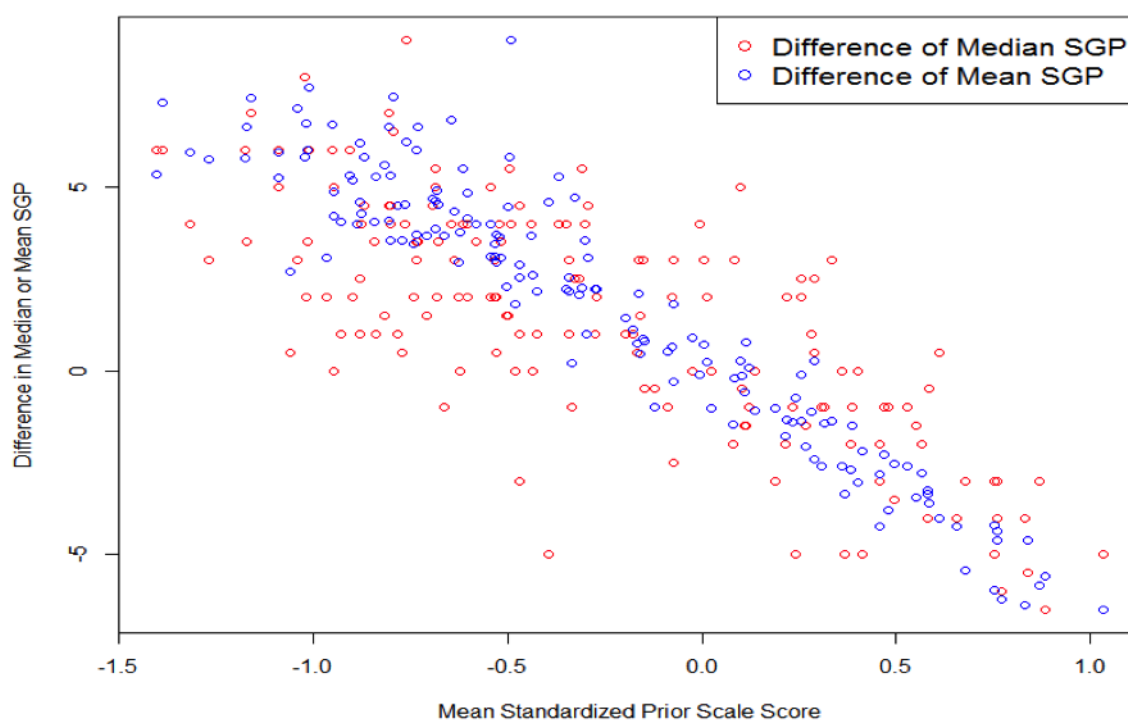


Figure 6: **Figure B.6** *SGP_SIMEX minus SGP Plotted against School Mean Standardized Prior Scores for American Literature*

5.3 *Can the SIMEX correction lower correlations between aggregated school SGP and school prior scores?*

Answer: Yes, SIMEX does reduce the correlations between aggregated SGP and prior mean scores at the school level (Tables B.14 - B.18). To be more specific:

- The magnitude of reduction ranges from 21% to close to 70%. The median magnitude of reduction is 38% for Median SGP and 40% for Mean SGP in READING, and 44% for Median and 38% for Mean in MATHEMATICS.
- SIMEX achieves this correlation reduction by generally raising the aggregated SGP for lower-performing schools and lowering the aggregated SGP for high-performing schools, as illustrated in Figure B.1.
- In a few cases, such as grades 5 and 7 for Science, and grades 5 and 8 for Social Studies, the correlations between SGP and prior scores are quite small to begin with, and SIMEX correction leads to negative correlations. In these cases, both the original and the corrected correlations are insignificant.

\caption{**Table B.14:** Reading*}

2012	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX
.....				
Gr 4, 1 prior	0.325	0.357	0.139	0.157
Gr 5, 1 prior	0.336	0.35	0.16	0.165
Gr 5, 2 priors	0.255	0.277	0.08	0.091
Gr 6, 1 prior	0.459	0.476	0.286	0.286
Gr 6, 2 priors	0.368	0.387	0.209	0.214
Gr 6, 3 priors	0.331	0.358	0.187	0.203
Gr 7, 1 prior	0.648	0.661	0.499	0.525
Gr 7, 2 priors	0.499	0.521	0.354	0.363
Gr 7, 3 priors	0.472	0.493	0.348	0.369
Gr 7, 4 priors	0.456	0.472	0.351	0.364
Gr 8, 1 prior	0.546	0.566	0.346	0.365
Gr 8, 2 priors	0.457	0.484	0.253	0.277
Gr 8, 5 priors	0.383	0.394	0.253	0.265
.....				

\caption{**Table B.15:** ELA*}

2012	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX
.....				

<i>2012</i>	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX
Gr 4, 1 prior	0.233	0.24	0.105	0.11
Gr 5, 2 priors	0.314	0.324	0.191	0.205
Gr 6, 2 priors	0.406	0.41	0.302	0.306
Gr 7, 2 priors	0.338	0.34	0.194	0.207
Gr 8, 2 priors	0.399	0.418	0.223	0.246
.....				

\caption{**Table B.16:** Mathematics*}

<i>2012</i>	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX
.....				
Gr 4, 1 prior	0.268	0.28	0.159	0.175
Gr 5, 1 prior	0.216	0.23	0.122	0.139
Gr 5, 2 priors	0.176	0.193	0.098	0.119
Gr 6, 1 prior	0.208	0.225	0.088	0.099
Gr 6, 2 priors	0.156	0.168	0.063	0.071
Gr 7, 1 prior	0.298	0.293	0.161	0.164
Gr 7, 2 priors	0.296	0.29	0.187	0.184
Gr 8, 1 prior	0.309	0.319	0.215	0.229
Gr 8, 2 priors	0.272	0.283	0.196	0.212
.....				

\caption{**Table B.17:** Science*}

<i>2012</i>	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX
.....				
Gr 4, 1 prior	0.269	0.294	0.14	0.164
Gr 5, 2 priors	0.064	0.076	-0.033	-0.029
Gr 6, 2 priors	0.2	0.221	0.12	0.14
Gr 7, 2 priors	0.05	0.068	-0.06	-0.042
Gr 8, 2 priors	0.277	0.286	0.205	0.215
.....				

\caption{**Table B.18:** Social Studies*}

<i>2012</i>	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX
.....				
Gr 4, 1 prior	0.24	0.252	0.147	0.154
Gr 5, 2 priors	0.041	0.059	-0.054	-0.039
Gr 6, 2 priors	0.132	0.152	0.071	0.091
Gr 7, 2 priors	0.196	0.222	0.117	0.136
Gr 8, 2 priors	0.007	0.013	-0.059	-0.052
.....				

5.4 *What about the correlations between Grand Aggregated SGP and Prior Grand Mean School Scale Scores?*

Answer: Tables B.19 and B.20 show that SIMEX does reduce correlation at the aggregate level across grades and subjects. The magnitude of reduction is about 1/5 to 1/3 depending on how many subjects are included.

\caption{**Table B.19:** Reading and Mathematics*}

<i>2012</i>	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX	School Count
.....					
Grs 4-5, priors 1-2	0.449	0.466	0.297	0.314	1,266
.....					

\caption{**Table B.20:** Reading, ELA, Mathematics, Science and Social Studies*}

<i>2012</i>	Median SGP	Mean SGP	Median SIMEX	Mean SIMEX	School Count
.....					
Grs 4-8, priors 1-2	0.576	0.586	0.447	0.463	1,792
.....					

5.5 *How stable are the SIMEX results?*

Answer: The stability depends on the methods of aggregation, school size, and number of SIMEX iterations. Specifically,

- With 20 SIMEX iterations, school Median SGP_SIMEX correlate with each other at above 0.95.
- With 20 SIMEX iterations, school Mean SGP_SIMEX correlate with each other at above 0.99.

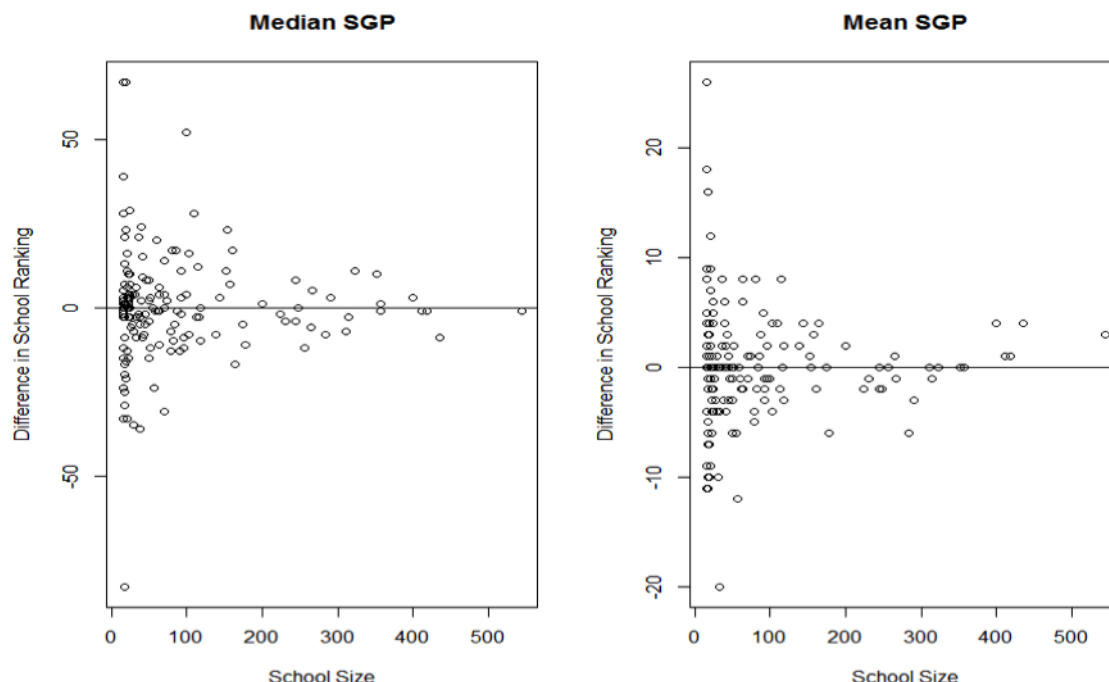


Figure 7: **Figure B.7** *Difference in School Ranking in Different SIMEX Trials Plotted Against School Size for American Literature*

- Because many schools are very close in terms of their median or mean SGP, we could see rather large difference in school rankings even if the difference in MGP is small, especially when school size is small, as illustrated in Figure 2.
- Figure 2 also shows that Mean SGP_SIMEX is much more stable than Median SGP_SIMEX.

5.6 *What are the consequences of including more priors?*

- Without SIMEX correction, including more priors leads to the mitigation of correlations between aggregated SGP and prior mean scores. The pattern is obvious and consistent in Tables B.14 - B.18, especially Table B.14, since it includes the same grade with different numbers of priors.
- Including more priors generally makes the SIMEX correction more powerful. The pattern is not always consistent (Tables B.14 - B.18).

5.7 *What about Goodness-of-Fit?*

The uncorrected SGP generally has better fit than the SIMEX corrected SGP. This is expected because goodness-of-fit means the extent to which the model fits the observed data, which is error-prone. Since SIMEX aims to adjust for measurement error, it naturally deviates from the observed data to a limited degree. Our confidence that SIMEX corrected SGP gets

closer to the “true” results comes from the outcome of the simulation studies presented in the second section of this report.

6 SIMEX implementation in the SGP package

The **SGP** package (Betebenner, VanIwaarden, Domingue & Shang, 2014) allows the user to specify many of the parameters used in the production of SIMEX SGP estimates. The `calculate.simex` argument of the `studentGrowthPercentiles` function requires the user to specify the following SIMEX parameters in a list with the following named elements:

- `state` identifies the two letter state abbreviation under which the test specific CSEMs are located in `SGPstateData`,
- `lambda` and `simulation.iterations` specify the desired values of λ and B respectively, and
- `extrapolation` to select a “linear” or “quadratic” extrapolant function.

The user may also request optional functionality, including

- `simex.sample.size` to specify a sample size of the data to be used in the production of the coefficient matrices¹,
- `save.matrices` to choose to save the coefficient matrices produced during each simulation experiment (TRUE or left NULL if not desired), and
- `simex.use.my.coefficient.matrices` to use previously computed coefficient matrices, if available (TRUE or left NULL if not), to produce fitted value estimates.

When the `calculate.simex` argument is TRUE in the high-level function `analyzeSGP` (rather than providing a list as described above) the package defaults are used. These defaults are to set λ to 0.5, 1, 1.5, 2 and B as 75, the sample size is set at 5,000, and the linear extrapolant is used. When computing cohort referenced SIMEX SGPs new coefficient matrices will be produced, used and saved. Previously computed coefficient matrices are used for baseline referenced SGPs (see the section below regarding SIMEX baseline matrix construction). These defaults were used in producing all Georgia’s SIMEX corrected SGPs.

Internally, the `studentGrowthPercentiles` function first uses the “naive” coefficient matrices (either calculated previously or during the current run) to obtain the “naive” fitted values from the unperturbed observed test scores. The (non-zero) values of λ are then iterated over, simulating B new data sets from the observed values each time. New coefficient matrices are produced if requested using each of the B data sets.² The function then finds the appropriate coefficient matrix that was either just produced, or is available in the `panel.data` list provided by the user or included in the package’s `SGPstateData`. Once the appropriate matrices have

¹Because the time taken to produce a coefficient matrix increases exponentially as the number of students increases a sample size smaller than the population can allow for satisfactory coefficient matrices to be produced in a more time efficient manner. When specified, the student population must be greater than the argument value. Note that the sample is only used to produce these matrices, and all students still receive SIMEX corrected SGP estimates.

²This includes producing the knots and boundaries used in the quantile regressions.

been located, the fitted value predictions at each percentile value are produced and then averaged over the B simulation iterations. Once these averages are obtained for each value of λ , the extrapolant function is applied to them to estimate the predicted value at $\lambda = -1$ is extrapolated for each student. These (extrapolated) predicted value estimates and the original observed scores are then used to produce SGP values in the typical manner.³

6.1 Baseline referenced coefficient matrix production

As with the construction of the “naive” baseline coefficient matrices, Georgia’s SIMEX adjusted coefficient matrices were produced using a “super-cohort” of students. That is, students with the same grade progression and/or course sequence from four or five academic year cohorts are combined into a single baseline norm group. The process of constructing the “naive” baseline coefficient matrices has been spread out over the years due to the availability of adequate prior test types for the various content areas, and care was taken to ensure that the same years’ data were used in the construction of the SIMEX matrices. For example, baseline referenced SGPs for Social Science were first produced in 2014 using data from 2010 through 2013 while the other CRCT subjects used data from 2008 to 2011 and *only* these same years were used to produce the SIMEX baseline matrices as well.

These combined cohort norm groups can range in size from several thousand in some of the less typical EOCT course progressions to well over 400,000 in the CRCT norm groups. Initial tests of the SIMEX SGP routine on the larger super-cohorts proved to be impossible in the R environment, causing numerous technical issues including total memory consumption and other memory related seg-faults. In order to deal with these issues, the ability to take a random sample of the student score histories for construction of the matrices was added to the SGP package. The sample size of 25,000 students was found to be adequate for providing consistent results, and the number of simulation iterations, B , was increased to 50 in order to compensate for added error from the sampling process (up from 30, the number identified earlier as the point at which the returns to reducing the MSE diminished).

The production of these SIMEX adjusted baseline matrices required several days to produce despite the use of parallel processing. After their construction, the matrices were added to the `SGPstateData` object and used to produce SIMEX adjusted SGPs for the 2011 - 2014 academic years.

7 Ranking SIMEX Corrected SGPs

Although SIMEX corrected SGPs are useful in reducing measurement error induced bias, they are not without technical limitations. At an individual level, the corrected SGPs have larger errors than the uncorrected, or “standard”, SGPs (McCaffrey, Castellano, & Lockwood, 2015; Shang et al., 2015). McCaffrey, et al. (2015) first suggested that ranking the SIMEX SGP values may present a possible alternative that would have the beneficial properties of both SGP estimate types. Castellano and McCaffrey (2017) recently investigated the properties of the percentile ranked SIMEX SGP (RS-SGP) at the aggregate level for MSGP estimates of

³From documentation sections not included here - the observed score is compared to all 100 predicted values. A students’ SGP is equal to the highest percentile at which the students observed score is greater than or equal to the corresponding predicted (fitted) value.

educator effectiveness. They found that the majority of the error variance in standard MSGP values is due to “sampling variability” (i.e. a classroom is considered a sample of all possible students), but that a substantial amount was also due to bias caused by ME. SIMEX correction can remove much of this bias initially. By subsequently taking the percentile ranks of the SIMEX corrected values and then aggregating these percentile ranks, the excess variance from the SIMEX estimation is removed. Furthermore they found that, at the individual level, the distribution of the RS-SGPs is also more uniformly distributed (similar to the standard SGP) rather than the SIMEX SGPs typical U-shaped distribution. The uniform distribution of the individual SGP values is a desirable characteristic because it suggests that the full range of SGP growth values (1-99) is equally likely to be attained.

Given the potential promise of RS-SGP, it is now calculated along with the SIMEX values in the **SGP** package⁴. Further insights from that implementation are discussed next.

7.1 SGP Package Implementation of Ranked SIMEX SGP

In their study, Castellano and McCaffrey report simply taking the percentile rank of the computed SIMEX SGP values to get the RS-SGP. However, unlike the SIMEX SGP values computed through data simulations in the **SGP** package, they compute their values using a closed-form equation. This produces continuous SIMEX SGP values, which allow for a more detailed ranking than using the integer values computed in the **SGP** package. Although their process helps to better understand the theoretical groundings of the various SGP estimates, it is only appropriate under particular assumptions about the data and ME structures that do not hold in the real-world situations.

Without a continuous value, the percentile ranking⁵ of a set of numbers that is already on a percentile scale does not produce results that differ substantially from the original in absolute value or distribution. Therefore a solution was required in the simulation process that would allow for a more continuous SIMEX SGP to be established. In following up with the authors they suggested that more granular SIMEX SGPs be established in the simulations, however this would require the already computationally and time intensive process to take 10 times longer. Furthermore, previously calculated SIMEX values would no longer be reproducible. A simpler solution was used that allows the estimated SIMEX values to be placed on a $1/8^{th}$ interval by calculating arithmetic midpoints between each percentile’s predicted score values⁶. This resulted in RS-SGP values that were more uniformly distributed in initial tests with real and simulated data.

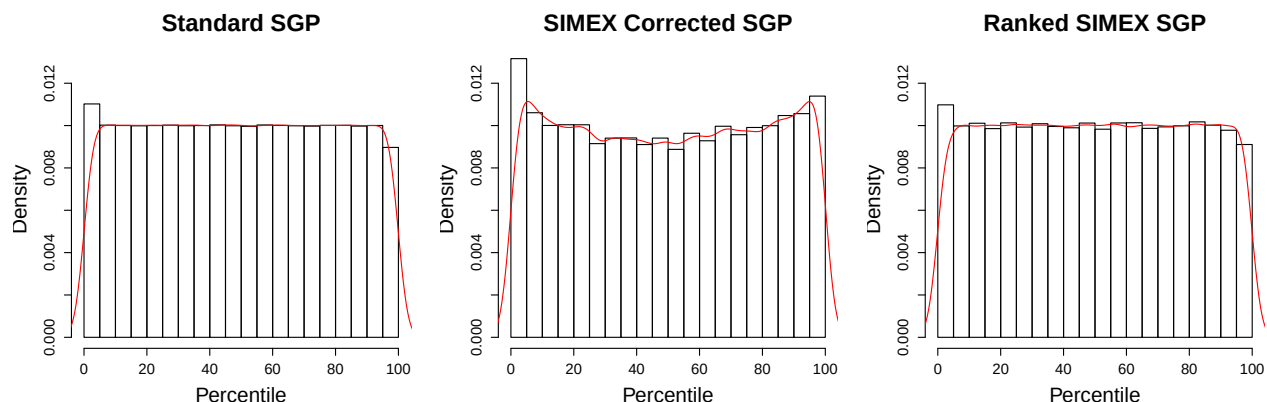
Figure 1 shows the distribution of the three types of SGP estimates for 2016 8th Grade Mathematics SGP analyses in Georgia: uncorrected (“standard”), SIMEX corrected and Ranked SIMEX. Note that these results are from analyses that use up to three years of data (two prior and the current year), which the authors indicate will also greatly reduce ME bias.

⁴SGP versions 1.7-0.0 and later

⁵Calculated as $(\text{rank}(\text{SIMEX SGP})/N) \times 100$ where N is the number of students. The result is rounded to the nearest integer.

⁶SGP estimates are found by predicting 100 scores for each student - one for each percentile. The position (1-99) of the predicted score that is closest to a student’s observed score is their estimated SGP.

Fig. B.1: Comparison of the Uniformity of Distributions for 8th Grade Mathematics Estimates.



By definition, the standard SGP is uniformly distributed *given any prior test score*, suggesting that any level of growth is equally likely regardless of prior achievement. This is a critical distinction, and Castellano and McCaffrey do not discuss the conditional uniformity of the RS-SGP. We find that this uniformity is not met in either the application of the closed-form equations to simulated data or in our initial tests with real data in the **SGP** package, although the RS-SGP distribution is much closer to uniform than that of the SIMEX SGPs.

The following figures are “Goodness of Fit” charts that are produced using the **SGP** package for each of the three SGP estimate types, and they can help investigate the SGP distribution in more detail. The “Student Growth Percentile Range” panel at bottom left shows the empirical distribution of SGPs given prior scale score deciles in the form of a 10 by 10 cell grid. Percentages of student growth percentiles between the 10th, 20th, 30th, 40th, 50th, 60th, 70th, 80th, and 90th percentiles were calculated based upon the empirical decile of the cohort’s prior year scaled score distribution. Perfect uniform distribution conditional on prior score would be indicated by a “10” in each cell. Deviations from perfect fit are indicated by red and blue shading. The further above 10 the darker the red, and the further below 10 the darker the blue. The bottom right panel of each plot is a **Q-Q plot** which compares the observed distribution of SGPs with the theoretical (uniform) distribution. An ideal plot here will show black step function lines that do not deviate from the ideal, red line which traces the 45 degree angle of perfect fit (as is seen here in the first plot for the standard SGP).

These plots display typical distributions of each SGP variant from the same 2016 8th Grade Mathematics SGP analyses as depicted above. The Standard SGPs are nearly perfectly distributed conditional upon prior achievement. The SIMEX and, to a lesser extent, RS-SGP distributions are skewed towards higher percentiles at the lower levels of achievement and lower growth for the higher prior achievement deciles.

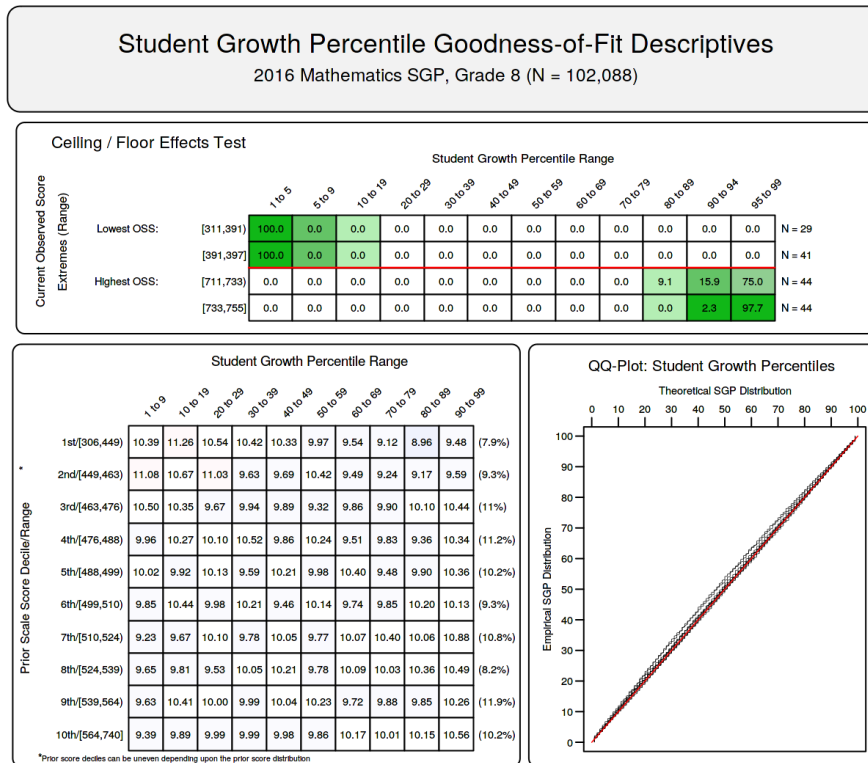
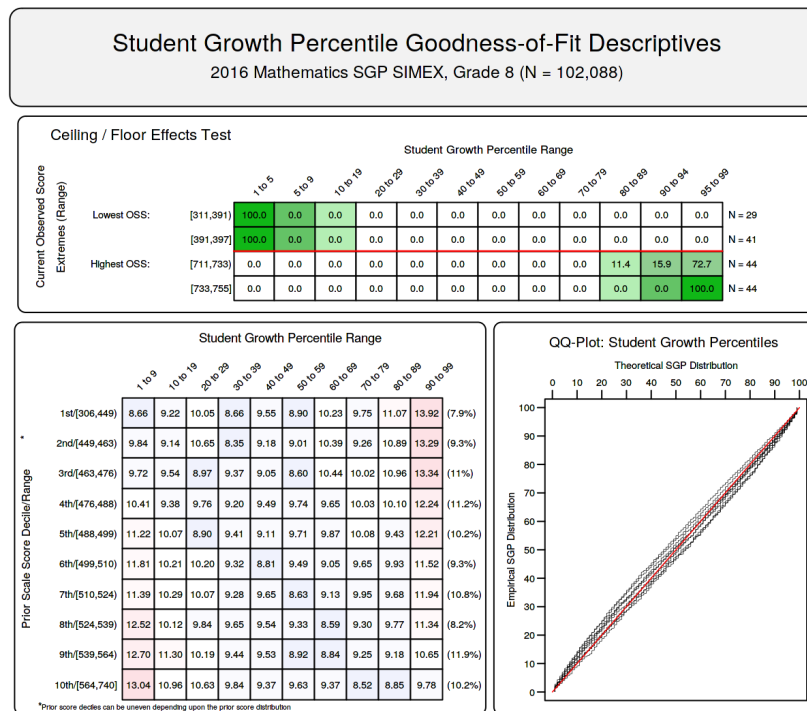
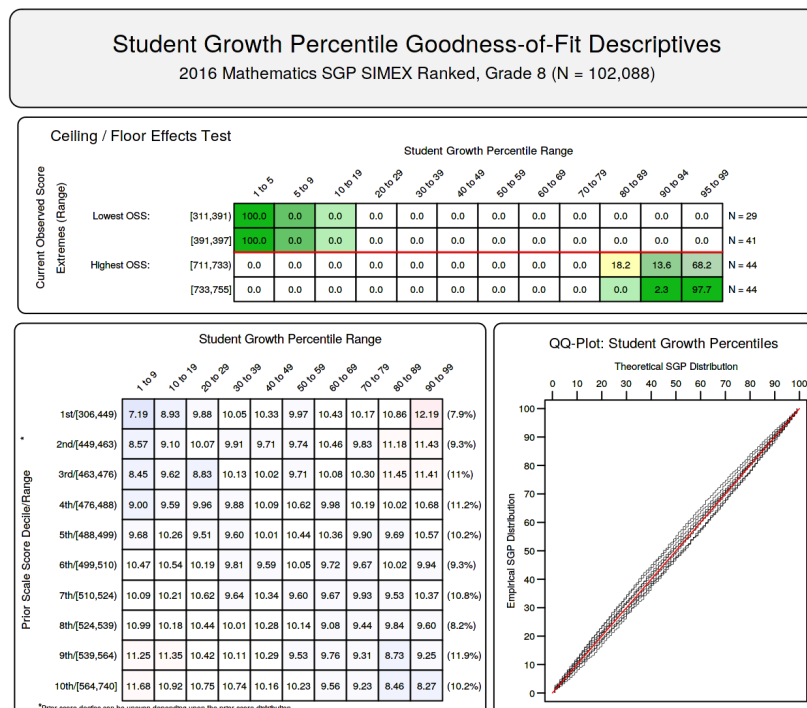
Fig. B.2: Goodness of Fit Plot for 2016 *Standard* 8th Grade Mathematics SGPs.

Fig. B.3: Goodness of Fit Plot for 2016 **SIMEX** 8th Grade Mathematics SGPs.Fig. B.4: Goodness of Fit Plot for 2016 **Ranked SIMEX** 8th Grade Mathematics SGPs.

8 Relationship of Ranked SIMEX with Prior Student Achievement

An important consequence of the typical SIMEX and RS-SGP conditional distributions is that a negative correlation is created between them and prior test scores. This is true at the student level and also translates to school and teacher aggregations. These negative relationships are indicative of the reduction in ME induced bias. The following tables show the results for the End-of-Grade (EOG) and End of Course (EOC) analyses for Georgia from 2016. As can be seen, the SIMEX and RS-SGP correlations are nearly identical in both settings. This is unsurprising as the maximum differences between the two values are between -3 and 3 with near-zero averages for all grade-by-subject specific analyses (see subsequent sections for more details on the observed differences between SIMEX and RS-SGP).

8.1 EOG Subjects

Table 1: Student Level Correlations between Prior Standardized Scale Score and 1) Current Scale Score, 2) Standard SGP, 3) SIMEX SGP and 4) Ranked SIMEX SGP.

Content Area	Grade	$r_{TestScores}$	$r_{Standard}$	r_{SIMEX}	r_{Ranked}	N Size
ELA	4	0.83	0.00	-0.12	-0.12	123,767
	5	0.83	0.01	-0.07	-0.07	122,053
	6	0.84	0.01	-0.07	-0.07	119,853
	7	0.84	0.02	-0.07	-0.07	118,962
	8	0.83	0.02	-0.07	-0.07	120,474
Mathematics	4	0.84	0.00	-0.10	-0.10	124,313
	5	0.84	0.01	-0.07	-0.07	122,680
	6	0.84	0.01	-0.06	-0.06	120,376
	7	0.87	0.02	-0.07	-0.07	118,986
	8	0.81	0.02	-0.06	-0.06	102,088
Science	4	0.79	0.00	-0.10	-0.10	124,092
	5	0.80	0.00	-0.07	-0.07	122,488
	6	0.80	0.00	-0.08	-0.08	120,164
	7	0.83	0.00	-0.07	-0.07	119,107
	8	0.78	0.00	-0.06	-0.06	94,694
Social Studies	4	0.77	0.00	-0.09	-0.09	123,446
	5	0.80	0.00	-0.08	-0.08	121,783
	6	0.80	0.00	-0.08	-0.08	119,550
	7	0.84	0.00	-0.08	-0.08	118,545
	8	0.82	0.00	-0.06	-0.06	119,813

8.2 EOC Subjects

Each EOC test subject is analyzed using more than one sequence of prior subjects, grades and years, and these unique progressions are disaggregated in Tables 2 and 3 using the most recent prior available for each norm group (although more prior years' scores are used in SGP calculations when available). These correlations between current and prior scale score are notably lower than in the grade level norm groups, and overall lower correlations may be expected in EOC subjects due to the change in specific subject from one course to the next.

As expected, the relationships between growth and prior achievement at the student level reported in Tables 2 and 3 are still non-existent for standard SGPs and slightly negative after SIMEX correction and SIMEX ranking. SIMEX and RS-SGP correlations are indistinguishable for EOCs as well.

Table 2: EOC Student Level Correlations between Prior Standardized Score and 1) Current Scale Score, 2) Standard SGP, 3) SIMEX SGP and 4) Ranked SIMEX SGP - Disaggregated by Norm Group.

Content Area	Most Recent Prior	$r_{TestScores}$	$r_{Standard}$	r_{SIMEX}	r_{Ranked}	N Size
Grade 9 Lit	2015 ELA Grade 7	0.75	0.02	-0.10	-0.10	4,075
	2015 ELA Grade 8	0.82	0.02	-0.07	-0.07	115,378
	2015 Grade 9 Lit	0.63	0.00	-0.09	-0.09	5,279
American Lit	2013 Grade 9 Lit	0.78	0.00	-0.10	-0.10	2,500
	2014 Grade 9 Lit	0.79	0.02	-0.07	-0.07	95,761
	2015 American Lit	0.65	0.00	-0.09	-0.09	4,596
	2015 Grade 9 Lit	0.77	-0.01	-0.10	-0.10	8,176
US History	2012 Soc Studies Grade 8	0.66	-0.01	-0.07	-0.07	1,909
	2013 Soc Studies Grade 8	0.72	0.00	-0.08	-0.08	84,784
	2014 Soc Studies Grade 8	0.74	0.00	-0.09	-0.09	10,283
	2015 Economics	0.76	-0.01	-0.10	-0.10	4,389
	2015 US History	0.67	0.00	-0.10	-0.10	2,590
	2016 Economics	0.72	0.00	-0.08	-0.08	2,081
Economics	2012 Soc Studies Grade 8	0.73	0.00	-0.08	-0.08	3,309
	2013 Soc Studies Grade 8	0.71	0.00	-0.08	-0.08	2,858
	2014 Soc Studies Grade 8	0.73	0.00	-0.11	-0.11	3,146
	2014 US History	0.70	0.00	-0.06	-0.06	5,832
	2015 Soc Studies Grade 8	0.74	0.00	-0.11	-0.11	2,225
	2015 US History	0.71	0.00	-0.08	-0.08	88,780
	2016 Economics	0.66	-0.01	-0.10	-0.10	1,518

Table 3: EOC Student Level Correlations between Prior Standardized Score and 1) Current Scale Score, 2) Standard SGP, 3) SIMEX SGP and 4) Ranked SIMEX SGP - by Norm Group (*Continued*).

Content Area	Most Recent Prior	$r_{TestScores}$	$r_{Standard}$	r_{SIMEX}	r_{Ranked}	N Size
Coordinate Algebra	2015 Coord Algebra	0.71	0.00	-0.11	-0.11	4,189
	2015 Math Grd 7	0.73	0.01	-0.10	-0.10	3,148
	2015 Math Grd 8	0.78	0.01	-0.07	-0.07	19,544
Analytic Geometry	2014 Coord Algebra	0.57	0.00	-0.08	-0.08	1,831
	2015 Analytic Geom	0.58	0.00	-0.10	-0.10	6,792
	2015 Coord Algebra	0.75	0.02	-0.07	-0.07	84,634
Algebra I	2015 Math Grd 7	0.76	0.00	-0.10	-0.10	15,646
	2015 Math Grd 8	0.80	0.02	-0.06	-0.06	75,621
Geometry	2015 Coord Algebra	0.75	0.02	-0.07	-0.07	33,006
Physical Science	2014 Biology	0.71	0.02	-0.05	-0.05	4,951
	2014 Science Grd 8	0.69	-0.01	-0.07	-0.07	3,236
	2015 Biology	0.73	-0.01	-0.08	-0.08	36,140
Biology	2015 Physical Science	0.58	0.00	-0.11	-0.11	2,944
	2015 Science Grd 7	0.73	0.00	-0.07	-0.07	26,138
	2015 Science Grd 8	0.78	0.00	-0.08	-0.08	18,411
	2014 Physical Science	0.84	0.00	-0.09	-0.09	2,753
	2014 Science Grd 8	0.74	-0.01	-0.07	-0.07	12,554
	2015 Biology	0.67	0.00	-0.10	-0.10	5,015
	2015 Physical Science	0.69	0.01	-0.09	-0.09	39,489
	2015 Science Grd 8	0.77	-0.03	-0.10	-0.10	60,608

8.3 Schools

We must also consider the impact the SIMEX ranking has on aggregated SGPs since they are used for school and teacher accountability in Georgia. The following tables look at the correlations between school level aggregations (averages) of prior student achievement and growth from analyses for Georgia. Three years of results are provided.

Table 4: 2014 to 2016 School Level EOG Correlations between Mean Prior Standardized Scale Score and Aggregate SGPs by Content Area.

Content Area	Year	R Mean SGP	R Mean SIMEX	R Mean Ranked SIMEX	N
ELA	2014	0.55	0.35	0.35	1,734
	2015	0.42	0.26	0.26	1,743
	2016	0.58	0.43	0.43	1,746
Mathematics	2014	0.45	0.32	0.32	1,734
	2015	0.46	0.36	0.36	1,743
	2016	0.51	0.41	0.41	1,746
Science	2014	0.41	0.25	0.25	1,737
	2015	0.40	0.28	0.28	1,746
	2016	0.50	0.39	0.39	1,745
Social Studies	2014	0.35	0.20	0.20	1,737
	2015	0.34	0.20	0.20	1,746
	2016	0.44	0.33	0.33	1,746

Table 5: 2014 to 2016 School Level EOC Correlations between Mean Prior Standardized Scale Score and Aggregate SGPs by Content Area.

Content Area	Year	R Mean SGP	R Mean SIMEX	R Mean Ranked SIMEX	N
Grade 9 Lit	2014	0.44	0.21	0.21	468
	2015	0.33	0.21	0.21	485
	2016	0.55	0.43	0.43	493
American Lit	2014	0.49	0.26	0.26	425
	2015	0.31	0.18	0.18	425
	2016	0.60	0.51	0.51	433
US History	2014	0.11	0.01	0.01	421
	2015	0.14	0.05	0.05	422
	2016	0.27	0.18	0.18	428
Economics	2014	0.15	0.06	0.06	416
	2015	0.12	0.04	0.04	424
	2016	0.38	0.29	0.29	425
Coordinate Algebra	2014	0.40	0.24	0.24	639
	2015	0.36	0.22	0.22	711
	2016	0.33	0.22	0.22	206
Analytic Geometry	2014	0.48	0.32	0.32	421
	2015	0.45	0.33	0.33	431
	2016	0.53	0.42	0.42	332
Algebra I	2016	0.35	0.24	0.24	560
Geometry	2016	0.59	0.48	0.48	122
Physical Science	2014	0.30	0.19	0.19	457
	2015	0.33	0.24	0.24	529
	2016	0.34	0.27	0.27	574
Biology	2014	0.21	0.08	0.08	426
	2015	0.30	0.20	0.20	436
	2016	0.33	0.23	0.23	438

9 Differences between SIMEX and Ranked SIMEX SGPs

Given that the correlations between prior achievement and growth are essentially identical for SIMEX and RS-SGPs, and yet there are visible differences in their distributions and model fit charts, we now investigate the absolute differences between the two values at the student and school levels.

9.1 SGP Change at Individual Student Level

When replacing SIMEX with ranked SIMEX, students with higher SIMEX SGP tend to receive lower ranked SIMEX SGP, students with lower SIMEX SGP tend to receive higher ranked SIMEX SGP. The correlation between SGP difference and SIMEX SGP is -0.82. Most of the students have SGP change within 2 percentiles.

Table 6: The Change of SGP (Ranked SIMEX - SIMEX) by EOG Subject and MeanGP SIMEX.

Subject	SGP Difference	Mean SIMEX SGP	N
ELA	-3	84.56	4,903
	-2	82.53	115,599
	-1	74.94	129,685
	0	50.71	93,362
	1	25.93	124,764
	2	18.52	125,827
	3	15.09	10,969
Mathematics	-4	85.00	19
	-3	82.50	8,834
	-2	81.21	116,896
	-1	73.41	132,958
	0	46.01	103,661
	1	22.51	152,706
	2	16.93	72,888
Science	3	19.45	481
	-3	82.27	974
	-2	81.08	87,952
	-1	75.75	153,206
	0	47.13	117,345
	1	22.71	160,754
	2	17.95	59,845
Social Studies	3	17.57	469
	-3	78.44	590
	-2	80.80	95,647
	-1	75.55	160,955
	0	46.18	125,384
	1	21.93	167,644
	2	16.95	52,626
	3	17.57	291

Table 7: The Change of SGP (Ranked SIMEX - SIMEX) by EOC Subject and MeanGP SIMEX.

Subject	SGP Difference	Mean SIMEX SGP	N
Grade 9 Lit	-3	84.19	1,229
	-2	83.39	24,230
	-1	75.51	23,621
	0	51.63	21,654
	1	25.16	26,820
	2	18.56	26,322
	3	16.40	856
American Lit	-5	95.00	6
	-4	90.36	184
	-3	84.93	3,206
	-2	82.28	26,494
	-1	70.17	19,128
	0	47.46	17,546
	1	23.81	27,111
	2	16.51	16,456
	3	14.97	891
	4	15.00	11
US History	-3	79.48	42
	-2	81.69	8,304
	-1	76.15	34,751
	0	43.85	29,461
	1	21.07	30,119
	2	19.75	3,351
	3	34.00	8
Economics	-4	70.44	16
	-3	76.48	182
	-2	79.48	4,256
	-1	75.06	39,196
	0	43.38	34,855
	1	20.38	28,252
	2	19.81	911

Table 8: The Change of SGP (Ranked SIMEX - SIMEX) by EOC Subject and MeanGP SIMEX (Continued).

Subject	SGP Difference	Mean SIMEX SGP	N
Coordinate Algebra	-3	88.11	132
	-2	82.42	3,935
	-1	77.28	6,543
	0	47.85	5,603
	1	22.82	7,710
	2	17.66	2,913
	3	20.76	45
Analytic Geometry	-3	82.38	945
	-2	82.56	13,398
	-1	76.40	22,639
	0	46.66	21,341
	1	22.32	26,164
	2	17.29	8,645
	3	16.11	125
Algebra I	-4	79.00	4
	-3	84.74	310
	-2	83.60	13,400
	-1	75.83	23,323
	0	46.04	19,656
	1	22.45	25,781
	2	18.07	8,790
	3	21.33	3
Geometry	-4	87.79	24
	-3	84.23	1,120
	-2	80.91	5,696
	-1	74.26	7,190
	0	50.15	5,323
	1	22.73	8,659
	2	17.73	4,436
	3	13.80	558

Table 9: The Change of SGP (Ranked SIMEX - SIMEX) by EOC Subject and MeanGP SIMEX (Continued).

Subject	SGP Difference	Mean SIMEX SGP	N
Physical Science	-3	82.81	16
	-2	82.45	5,699
	-1	76.95	29,913
	0	45.79	24,084
	1	21.80	30,114
	2	19.10	1,991
	3	19.00	3
Biology	-3	85.43	606
	-2	80.74	22,085
	-1	75.25	26,934
	0	48.13	21,032
	1	23.12	29,226
	2	18.53	20,181
	3	16.25	355

9.2 Mean SGP Change at School Level

Most of the schools have mean student growth percentile (MSGP) change within 2 percentiles when switching from SIMEX to Ranked SIMEX.

Table 10: Descriptive statistics of school MGP change to ranked SIMEX SGP (Ranked - SIMEX).

Subject	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	SD	N
ELA	-0.67	-0.09	0.06	0.07	0.22	1.56	0.24	1,746
Mathematics	-1.29	-0.36	-0.16	-0.14	0.07	1.17	0.31	1,746
Science	-1.06	-0.21	-0.06	-0.05	0.11	0.81	0.24	1,745
Social Studies	-1.03	-0.29	-0.11	-0.11	0.06	0.97	0.27	1,746
Grade 9 Lit	-1.16	-0.14	0.05	0.04	0.24	1.06	0.32	493
American Lit	-1.06	-0.34	-0.13	-0.10	0.13	1.31	0.35	433
US History	-0.72	-0.27	-0.09	-0.08	0.12	0.93	0.29	428
Economics	-0.80	-0.31	-0.14	-0.13	0.01	0.60	0.23	425
Coordinate Algebra	-1.39	-0.39	-0.03	-0.08	0.26	0.82	0.48	206
Analytic Geometry	-0.97	-0.25	-0.04	-0.05	0.15	0.71	0.30	332
Algebra I	-1.41	-0.44	-0.13	-0.14	0.19	0.90	0.45	560
Geometry	-1.04	-0.27	0.07	0.00	0.26	0.89	0.38	122
Physical Science	-1.11	-0.28	-0.06	-0.08	0.15	0.67	0.33	574
Biology	-0.93	-0.24	0.01	0.00	0.20	1.06	0.35	438

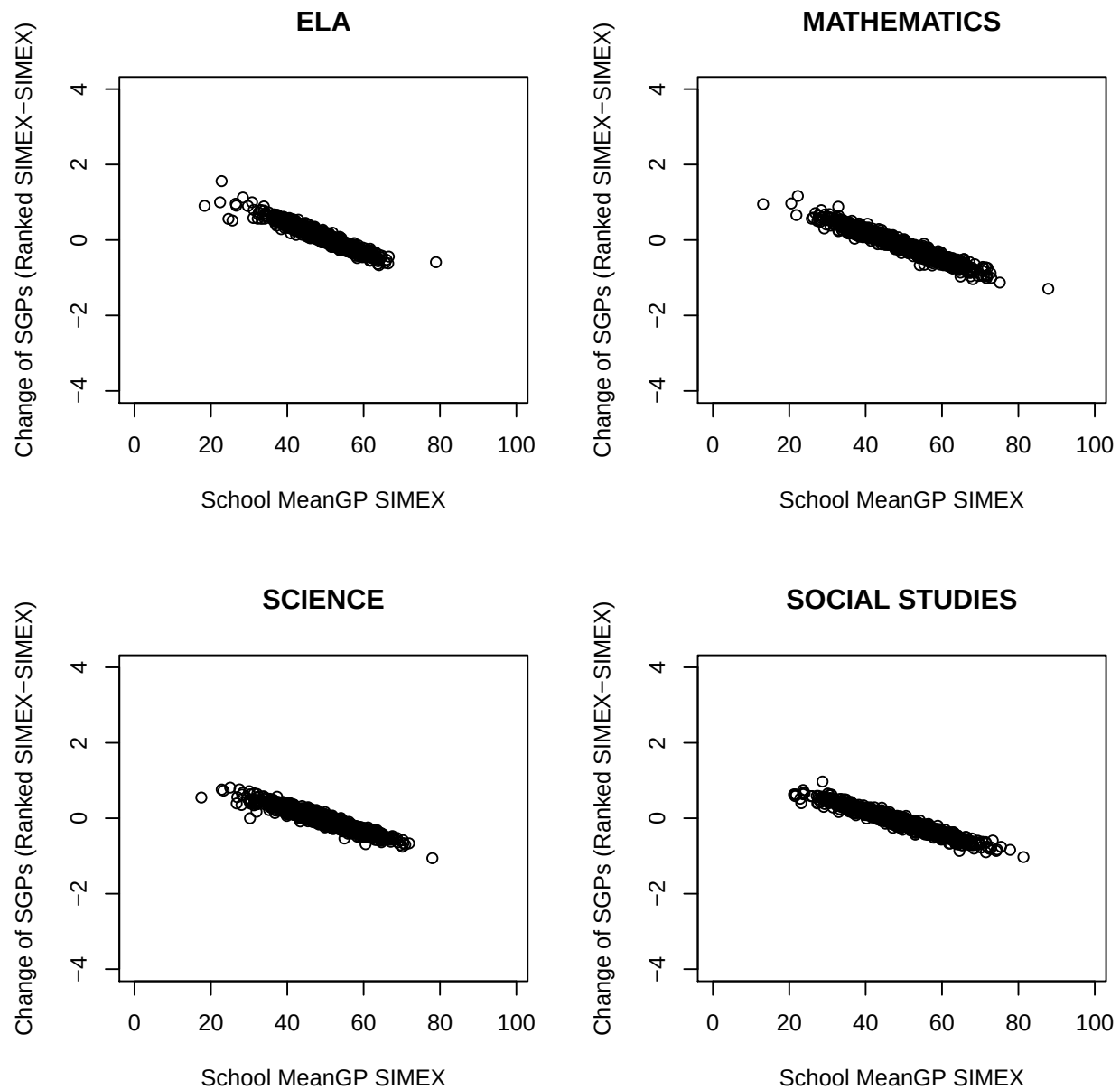
Fig. B.5: School MGP change (MGP Ranked SIMEX - MGP SIMEX) by EOG Subjects.

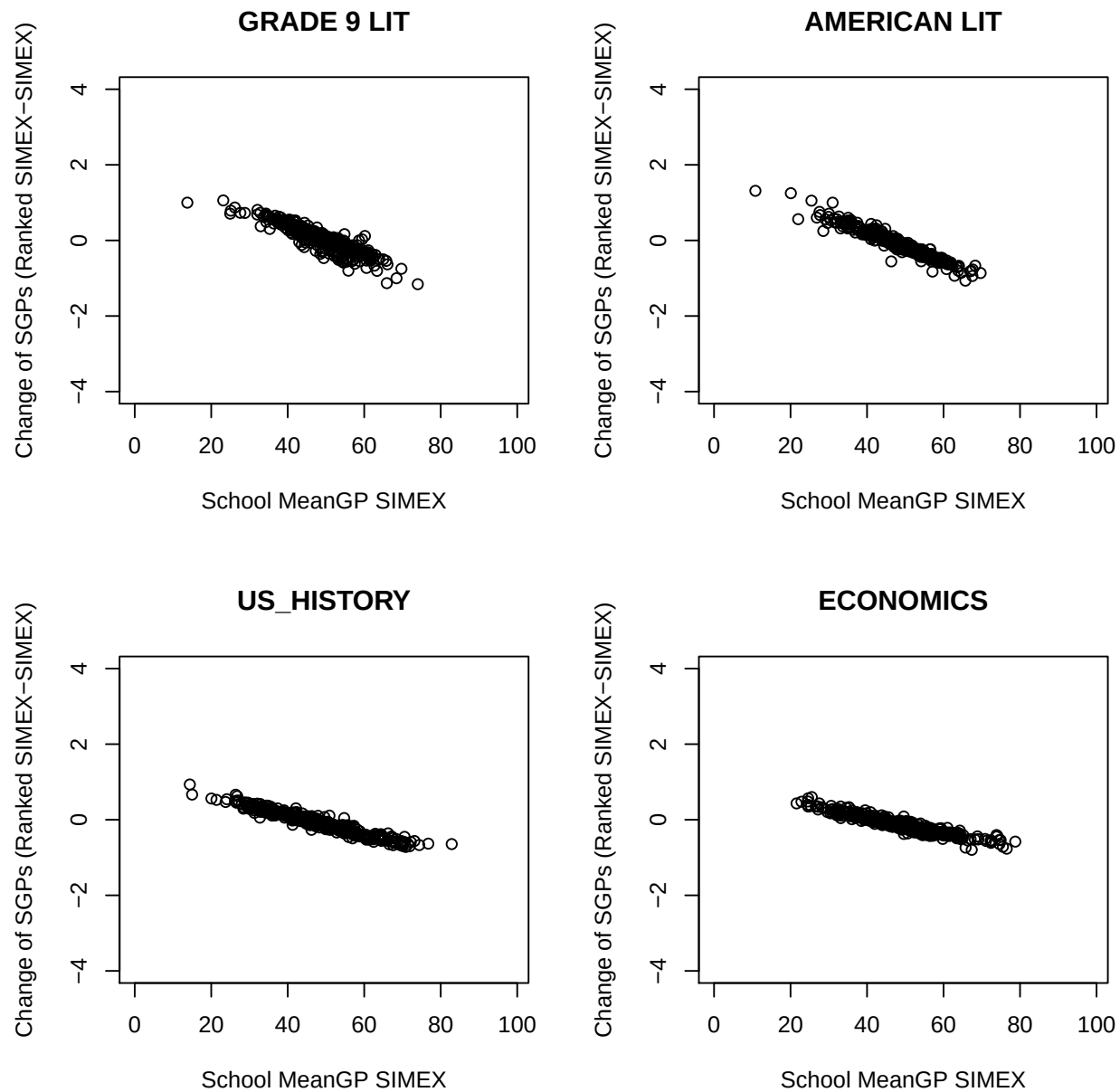
Fig. B.6: School MGP change (MGP Ranked SIMEX - MGP SIMEX) by EOC Subjects.

Fig. B.7: School MGP change (MGP Ranked SIMEX - MGP SIMEX) by EOC Subjects (Continued).

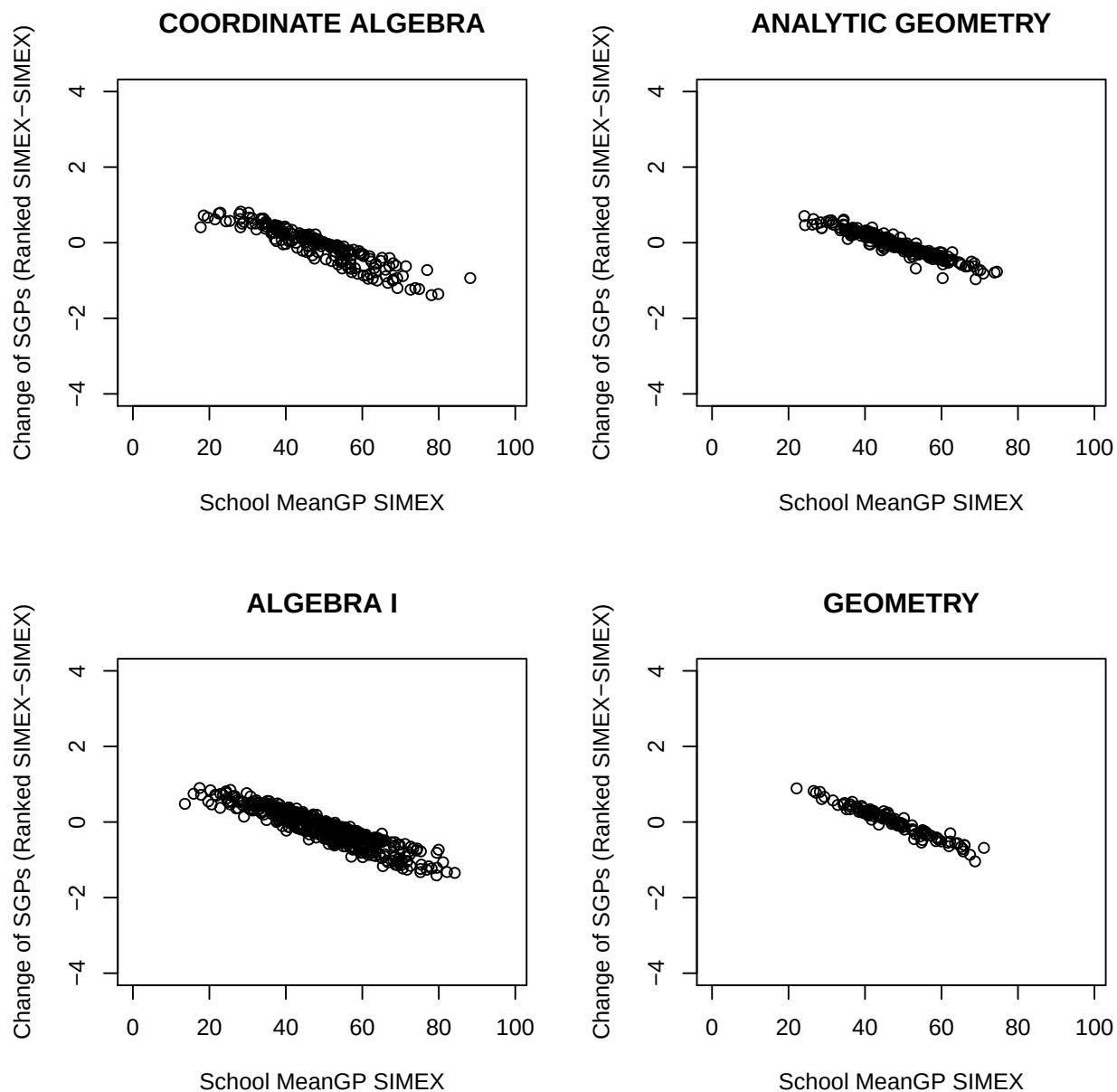
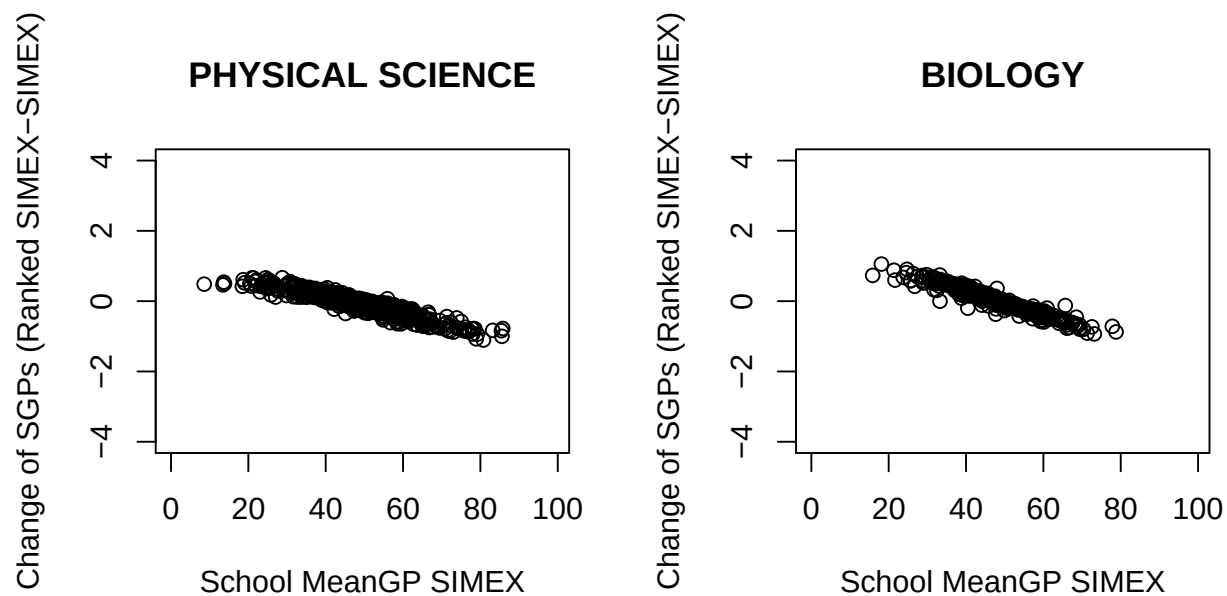


Fig. B.8: School MGP change (MGP Ranked SIMEX - MGP SIMEX) by EOC Subjects (Continued).



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